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Characterizing stable regions in the residual stream of LLMs

Jett Janiak
LASR Labs
jettjaniak@gmail.com

Jacek Karwowski*
University of Oxford
jacek.karwowski@cs.ox.ac.uk

Chatrik Singh Mangat*
LASR Labs
chatrikmangat@outlook.com

Giorgi Giglemiani
LASR Labs
giglemiani@gmail.com

Nora Petrova
LASR Labs
nora.axion@gmail.com

Stefan Heimersheim
Apollo Research
stefan@apolloresearch.ai

Abstract

We identify *stable regions* in the residual stream of Transformers, where the model's output remains insensitive to small activation changes, but exhibits high sensitivity at region boundaries. These regions emerge during training and become more defined as training progresses or model size increases. The regions appear to be much larger than previously studied polytopes. Our analysis suggests that these stable regions align with semantic distinctions, where similar prompts cluster within regions, and activations from the same region lead to similar next token predictions. This work provides a promising research direction for understanding the complexity of neural networks, shedding light on training dynamics, and advancing interpretability.

1 Introduction

We study the effects of perturbing Transformer activations, building upon previous work [6, 9, 12]. Specifically, we interpolate between different residual stream activations after the first layer, and measure the change in the model output. Our initial results suggest that:

1. The residual stream of a trained Transformer can be divided into *stable regions*. Within these regions, small changes in model activations lead to minimal changes in output. However, at region boundaries, small changes can lead to significant output differences.
2. These regions emerge during training and evolve with model scale. Randomly initialized models do not exhibit these stable regions, but as training progresses or model size increases, the boundaries between regions become sharper.
3. These stable regions appear to correspond to *semantic distinctions*. Dissimilar prompts occupy different regions, and activations from different regions produce different next token predictions.
4. These stable regions appear to be much larger than previously studied polytopes [2, 7, 13]. Our analysis of gate activations shows that while interpolating between two prompts typically crosses only one stable region boundary, hundreds or thousands of gates change their activation sign (Appendix F).

*equal contribution

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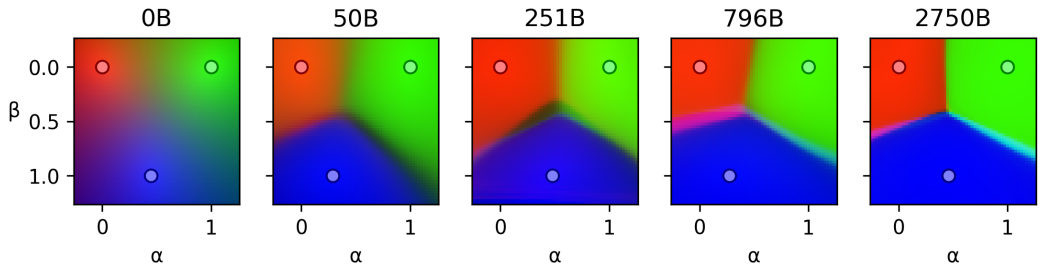

Figure 1: Visualization of stable regions in `OLMo-7B` during training. Colors represent the similarity of model outputs to those produced by three model-generated activations (red, green, blue circles). Each subplot shows a 2D slice of the residual stream after the first layer at different stages of training, with the number of processed tokens indicated in the titles. As training progresses from left to right, distinct regions of solid color emerge and the boundaries between them sharpen. Refer to the end of Section 4 for details.

2 Related work

Previous work on characterizing the complexity of neural networks includes studying linear regions (aka polytopes) in piecewise linear networks [2, 7, 13] and bounding the VC dimension of network architectures [1, 8]. While these approaches provide theoretical insights, the extent to which they describe the practical expressivity of networks remains unclear. In this work, we study stable regions in the residual stream of Transformers that can be mapped to classes of very similar outputs. Our initial results suggest that stable regions are much larger than previously studies polytopes.

The linear representation hypothesis [3, 15, 16] suggests that features in neural networks correspond to directions in activation space. Since residual stream activations can be though of as sums of features, we can similarly think about model-generated activations as vectors. Due to normalization layers in Transformers, these vectors lie closely to a hypersphere in the residual stream. Because of high dimensionality of this space, most linear interpolations between two activations will stay close to the hypersphere.

Heimersheim & Mendel [9], motivated by theoretical results by Hänni et al. [10], studied robustness to noise in GPT-2 by interpolating between two residual stream activations and they observed nonlinear changes in the model output. Follow up studies by Giglemiani et al. [4], Lee & Heimersheim [11] connect this to SAEs work by Gurnee [6], Lindsey [12].

3 Methods

For a given prompt p, we define a Transformer forward pass with activation patching as a function $F_p : \mathbb{R}^D \to \mathbb{R}^D$ that returns residual stream activations after last layer[1], at last sequence position. Specifically, $F_p(X)$ replaces the residual stream activations after the first[2] layer, at the last sequence position, with X, before applying the rest of the model layers.

If small changes in model activations lead to minimal changes in output, then the residual stream after the first layer can be divided into stable regions. In this case, F_p's rate of change should be small away from region boundaries and large when approaching and crossing region boundaries.

To study this efficiently, we interpolate between prompts p_A and p_B, which may or may not belong to the same region. We introduce an interpolation coefficient $\alpha \in [0, 1]$, where $\alpha = 0$ corresponds to prompt p_A and $\alpha = 1$ corresponds to prompt p_B.

We measure the L_2 distance $d(\alpha) : [0, 1] \to \mathbb{R}$ between two outputs: a clean run on prompt p_A and a patched run where activations are modified depending on α. In the patched run, we replace the activations with a linear interpolation $A + \alpha(B - A)$, where A and B represent the model-generated activations after the first layer, at the last sequence position, corresponding to prompts p_A and p_B

[1] Before the unembedding and the softmax.

[2] Preliminary results in Appendix E suggest that some of the reported effects hold for later layers.

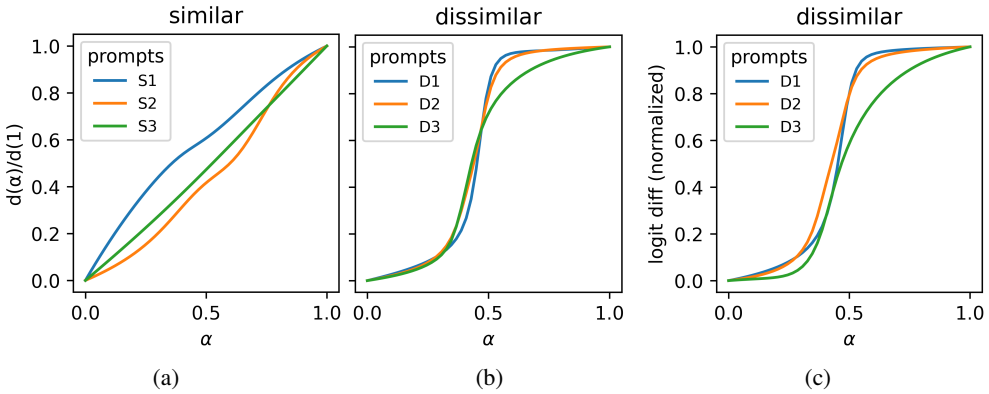

Figure 2: (a,b) Relative output distance as a function of α for (a) similar and (b) dissimilar pairs of prompts in `Qwen2-0.5B`. (c) Normalized logit difference between top prediction for p_B and p_A.

respectively. Formally, this distance is expressed as:

$$d(\alpha) = \|F_{p_A}(A) - F_{p_A}(A + \alpha(B - A))\|_2.$$

In our experiments, we use models from `Qwen2` [17] and `OLMo` [5] families with a number of parameters ranging from 0.5B to 7B, see Appendix A for details. We compute the distance d for 50 uniformly spaced values of α between 0 and 1.

4 Experiments

Illustrative examples We suspect that semantically similar prompts are more likely to belong to the same stable region than semantically different prompts. To test this, we construct pairs of prompts dissimilar to each other numbered D1-D3, and pairs of similar prompts numbered S1-S3. We present prompts D1 and S1 below, and the remaining prompts in Appendix B. For simplicity, the prompts differ only in the last token.
Dissimilar prompts D1:
p_A = " The house at the end of the street was very", top prediction = " quiet"
p_B = " The house at the end of the street was in", top prediction = " a"
Similar prompts S1:
p_A = " She opened the dusty book and a cloud of mist"
p_B = " She opened the dusty book and a cloud of dust"

In Figure 2 we plot the relative distance $\frac{d(\alpha)}{d(1)}$ in model outputs as we interpolate between two prompts in `Qwen2-0.5B`[3]. In Figure 2a, we show the results for interpolation between similar pairs S1-S3, and in Figure 2b for dissimilar pairs D1-D3. We observe that the shapes for similar prompts S1-S3 are close to linear, or equivalently that the rate of change is similar throughout the interpolation. This is in contrast to results for dissimilar pairs D1-D3, where we observe flat regions at the beginning and at the end of the interpolations, separated by a sharp jump, which is in line with what Heimersheim & Mendel [9] observed in `GPT-2`. This relationship between semantic similarity and stable regions is further supported by a large-scale analysis of prompt pairs filtered by their maximum sensitivity (Appendix G). Our interpretation of these results is:

1. Activations for dissimilar prompts appear to occupy distinct stable regions. This is evidenced by the non-linear shape of $d(\alpha)$, which shows flat areas near the model-generated activations and a sharp jump between them.
2. Activations for similar prompts likely belong to the same stable region, as indicated by the linear change in $d(\alpha)$..

[3]For bigger models from the Qwen2 family, we observe smaller difference between semantically similar and different prompts. We hypothesize that more capable models are able to make more precise predictions, even for very similar prompts.